

SHRUTI ROHOM

Computer Engineer

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Kopargaon, 423603

PROFILE SUMMARY

Computer Engineering student with strong knowledge of Python, Artificial Intelligence, Machine Learning, databases, and problem-solving. Experienced in developing AI/ML applications and data-driven projects using Python, TensorFlow, Scikit-learn, and REST APIs. Skilled in machine learning model development, data preprocessing, and deep learning. Eager to apply AI/ML skills and contribute to real-world software and intelligent systems.

EDUCATION

Sanjivani College Of Engineering, Kopargaon

B.Tech in Computer Engineering | 2023-2027 | CGPA - 9.02

Higher Secondary (HSC), Science (PCMB) | 2022-2023

Sanjivani Junior College, Kopargaon | Percentage - 74.33%

Secondary School (SSC) | 2021

Percentage - 95.20%

CORE SUBJECTS AND CONCEPTS

- Programming Languages: Python, Java, C++
- Web Technologies: HTML, CSS, JavaScript, PHP, Bootstrap, REST APIs
- Machine Learning & Deep Learning: TensorFlow, Scikit-learn, EfficientNetB0, OpenCV, Recommendation Systems
- OOP (Object-Oriented Programming) - Classes, Inheritance, Polymorphism, Encapsulation
- IoT & Communication: MQTT Protocol, ThingSpeak
- DSA (Data Structures & Algorithms) - Arrays, Linked Lists, Trees, Graphs, Searching & Sorting
- Database - SQL, MySQL, MongoDB, CRUD operations
- Tools & Platforms: GitHub, VS Code, Eclipse, Jupyter Notebook, ThingSpeak (IoT projects)

PROJECTS

1. AI-Based Plant Disease Detection & Treatment Recommendation System

- Developed a deep learning system to classify tomato plant diseases from leaf images with 92.56% validation accuracy.
- Implemented EfficientNetB0, image preprocessing, confidence scoring, and treatment recommendations for disease identification.
- Technologies: Python, TensorFlow, EfficientNetB0, OpenCV, FastAPI, React, Tailwind CSS**

2. E-Commerce Recommendation System

- Developed an AI-based recommendation system for suggesting products based on user preferences and browsing history.
- Implemented content-based and collaborative filtering techniques to provide personalized product recommendations and improve user experience.
- Technologies: Python, Pandas, NumPy, Scikit-learn, Machine Learning, Collaborative Filtering, Content-Based Filtering, Matplotlib, Jupyter Notebook**

3. Opticals E-Commerce Website

- Developed a responsive website for an optical store to showcase and sell eyewear online.
- Included product listing, categories, shopping cart, and checkout functionality.
- Technologies: HTML, CSS, JavaScript, Bootstrap, Font Awesome, PHP/MySQL**

4. Mutual Fund Recommendation System

- Developed a Python-based recommendation system to suggest suitable mutual funds based on user preferences and fund performance.
- Implemented data preprocessing, filtering, and recommendation logic to compare mutual funds using parameters such as returns, risk, and investment preferences.
- Provided personalized fund recommendations to help users make informed investment decisions.
- Technologies: Python, Pandas, NumPy, Scikit-learn, Machine Learning**

5. Footstep Power Generation Using Electromagnetic Induction

- Developed a footstep-based power generation system that converts mechanical energy from human footsteps into electrical energy using electromagnetic induction.
- Designed a mechanism using magnets and coils to generate electrical energy and store the generated power.
- Demonstrated an eco-friendly and renewable energy solution for low-power applications.
- Technologies: Electromagnetic Induction, Magnets & Coils, DC Circuit, Energy Storage**

ACHIEVEMENTS

- 5-Star Rating on HackerRank - DBMS, C, C++
- NPTEL Silver Certificate - Employment Communication
- Completed Certification in Data Science - Practical skills in Python, Pandas, NumPy, and data analysis
- Published Paper: "Breath Safe: An IoT-Based Air Quality Monitoring System" - MAT Journal, 2025
- Participated in Coding Competitions - Enhanced problem-solving and algorithm skills
- Participated in Shelnspires Hackathon (Zensar), a national-level AI hackathon focused on real-world problem solving and innovation